

Learning from Click-Logs: a Two-Stage Retrieve-then-Rank System for MIND and EB-NeRD

CS4.406 Information Retrieval & Extraction — Assignment 2 · Chaitanya Shah Done **solo**, which the specification permits for this team assignment.

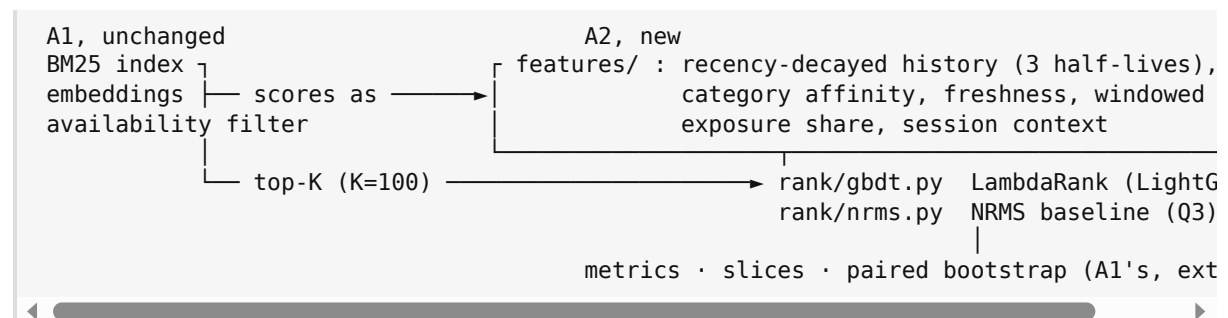
Code: https://github.com/ChaitanyaShah21/IRE_A2 · decision log:

ARCHITECTURE.md (A1's D1–D32, A2's D33–D41) · every quoted figure traces to reports/NUMBERS.md · 326 tests, including the Q9 leakage suite.

1. What was built, and the one idea behind it

A1 ended on a measured gap: **BM25 and embedding retrieval contain no time term** — a structural fact about both formulas — while **92.7% (MIND) / 93.5% (EB-NeRD) of real clicks are on fresh articles**. Three independent routes reached it, the sharpest being that train-window popularity predicts EB-NeRD clicks in the *wrong* direction (AUC 0.4647).

A2 acts on it. The A1 scorers are not replaced; **they become features** of a trained re-ranker that also sees time:



Because `cos_inf` (infinite half-life) reproduces A1's semantic score exactly and `bm25` is A1's lexical score, **"before re-ranking" is one column of "after"** — the Q2 comparison is two orderings of the same candidate list, not two unrelated systems.

Two candidate sets, deliberately kept apart (D37). Both leaderboards ask us to re-rank a *supplied* list (`article_ids_inview / impressions`); Q2 also asks us to *retrieve* top-K from the whole corpus. These are different operations on different candidate sets, and conflating them is the easiest way to report a meaningless number. Every table below says which it is.

2. Q1 — features, and the boundary that makes them honest

Feature	MIND	EB-NeRD
<code>cos_short / cos_medium / cos_inf</code> — decayed user profile · candidate (D35)	✓ (3/10/∞ positions)	✓ (1 d / 7 d / ∞)
<code>cat_share_short / medium / inf</code> — recency-weighted category affinity	✓	✓
<code>bm25</code> — A1's lexical score, N=100 (D31)	✓	✓
<code>freshness_hours</code> — $T - \min(\text{published_time}, \text{first_seen})$ (D34c)	✓ (first_seen only)	✓
<code>exposure_share_1h / _24h</code> — label-free share, strictly before T (D34b)	✓	✓
<code>hours_since_last_click</code> (D35)	—	✓
<code>session_position, minutes_since_session_start, device_type</code> (D34a)	—	✓

The feature set is an allowlist, never a denylist, and the builder *asserts* its output columns equal keys + label + that list. This is not tidiness: `total_pageviews` and `total_inviews` already sit in our `articles.parquet` and are whole-dataset aggregates containing the future, so any builder that loops over article columns picks them up and nothing errors.

Boundary enforcement (Q1.4, Q9.2). `tests/test_no_leakage.py` asserts two *properties* on real validation data for both datasets, rather than a list of per-feature checks: *future deletion* (delete every impression after τ ; features at or before τ must be byte-identical) and *label blindness* (shuffle the clicks; every feature must be identical). A new feature is covered automatically. The suite is mutation-verified: four realistic forward leaks were planted in production code and **all four were caught** — but only after extending the check to MIND, because on EB-NeRD `published_time` almost always wins the `min`, so the `first_seen` path is never exercised there.

Three landmines, each measured rather than assumed. EB-NeRD's catalogue runs a month past the test week (2,148 articles published after the train week ends), so **no catalogue-wide statistic is ever a feature**. Its `session_id` is shared by multiple users (1,032 ids), so sessions are keyed (**user, session**) — keyed by id alone the longest "session" is 14 days, keyed properly 29 minutes. And 1,810 candidate rows are shown *before* their `published_time`, which is why freshness takes the earliest evidence of existence.

3. Q2 — the trained re-ranker, before and after

Model (D36). LightGBM with the `lambdarank` objective: it is scored only on ordering each impression's clicked candidates above its unclicked ones, weighted by how much a swap would move nDCG. Trained on the train split, tree count chosen by validation nDCG@10, **every number below from the local test split**, which chose nothing. Hyperparameters were fixed before any result was seen. LightGBM learns impression boundaries from group *sizes* alone, so a shuffle trains happily and ranks garbage; `group_sizes` derives them from the ids and refuses non-contiguous input.

3.1 Supplied inview lists — what the leaderboards grade

System	MIND AUC	MIND nDCG@10	EB-NeRD AUC	EB-NeRD nDCG@10
random	0.4982	0.2756	0.5007	0.4288
bm25 (A1)	0.5641	0.3265	0.5041	0.4305
semantic (A1) — <i>before</i>	0.6097	0.3588	0.5457	0.4585
LambdaRank (A2) — <i>after</i>	0.6144	0.3636	0.7428	0.6250
paired gain over <i>before</i>	+0.0048 [0.0012, 0.0080]	+0.0048	+0.1971 [0.1953, 0.1989]	+0.1665

Both AUC gains exclude zero. **The asymmetry is the finding, not a disappointment.** Importances say why: on EB-NeRD freshness (38%) and exposure share (34%) carry 73% of the gain; on MIND `cos_inf` leads (27%) and the model saturates after 18 trees — validation nDCG@10 is flat from ~20 to 300 trees while training climbs 0.396 → 0.457. Single-feature AUCs on MIND put freshness at 0.534 and exposure at 0.533 against similarity's 0.610, because **MIND ships no publication times and no history timestamps**: its freshness is inferred from first appearance in a sampled log and its recency can only use

list position. The time signal helps where the data records time, which is also why the two datasets' decay features are not comparable to each other.

3.2 Retrieved top-K (K = 100) — our own candidate generation

A1's semantic generator retrieves 100 from the whole corpus; the same model re-ranks them (full numbers in `reports/retrieved_regime*_test_k100.csv`).

Recall@100 is 2.6% / 2.5%, so ranking metrics are defined on 798 / 4,704 impressions only — re-ranking cannot rescue an impression whose clicked article never entered the 100, which is why both leaderboards score a supplied list instead. Within those, MRR moves 0.070 → **0.260** (MIND) and 0.052 → **0.567** (EB-NeRD). EB-NeRD's AUC there (0.508 → 0.968) is **not** a better system than §3.1's 0.743: the 99 negatives were pulled from a 125,541-article corpus by text similarity, so most were never in circulation that day and freshness separates them trivially. Easy negatives inflate AUC; the supplied-list number is the one to believe.

4. Q3 — baseline reproduced, then beaten

Baseline (D39): NRMS (Wu et al., 2019) — multi-head self-attention over the user's clicked articles, then additive attention to pool them, scored by dot product against the candidate, trained with one click against four same-impression negatives. **One documented deviation:** the news encoder is a learned projection of our existing 384-dim multilingual sentence embeddings instead of word-level attention over title tokens. That is what makes it trainable on a GPU-less laptop; the user encoder, objective and scoring are the paper's. Both that substitution and D41's training budget push our absolute NRMS numbers below what a fully-trained, word-level NRMS reaches on a GPU — so this section is a *controlled ablation around a reproduced architecture*, not a claim about NRMS's ceiling.

The one principled change (D40): a time term. NRMS has no time input at all. A small MLP over (log freshness, exposure share 1 h, exposure share 24 h) is added to the click score, zero-initialised so training *starts* at exactly the baseline. The change was chosen from A1's measurements **before** any NRMS result existed — A1's Finding 5 is that selecting a claim after seeing its result is a tautology in disguise.

Ablation (Q3.3) and significance (Q3.4). Four arms, identical in everything but the time inputs, each trained on the same seeded 60,000 impressions (D41) and evaluated on the whole test split. Paired bootstrap, 1,000 resamples, differences against the baseline.

Arm	MIND AUC	paired Δ	EB-NeRD AUC	paired Δ
nrms (reproduced baseline)	0.6146	—	0.5874	—
+ freshness	0.6159	+0.0013 [0.0004, 0.0022]	0.7014	+0.1141 [0.1126, 0.1155]
+ exposure share	0.6122	−0.0024 [−0.0032, −0.0014]	0.6244	+0.0370 [0.0359, 0.0380]
+ both — the pre-registered change	0.6144	−0.0002 [−0.0011, 0.0008]	0.6738	+0.0864 [0.0852, 0.0877]

On EB-NeRD the claim holds with room to spare: +0.0864 AUC, CI clear of zero, and +0.0628 MRR. **On MIND it does not:** the pre-registered arm is indistinguishable from the baseline on AUC (−0.0002, interval straddling zero) though it does lift MRR (+0.0053 [0.0038, 0.0067]).

What the ablation buys over a single "improved" number: it **localises** the gain (freshness carries it; exposure alone adds +0.037 and *subtracts* when combined, the two being correlated — a fresh article is also a heavily-pushed one); it **reproduces the Q2 ordering in a different model class** (LambdaRank importances also rank freshness first on EB-NeRD and find exposure inverse on MIND), which is the kind of agreement that makes a finding believable; and it **says where the change does not transfer, and why** — MIND has no publication times and no history timestamps, so there is less time signal to add.

Two limits we state rather than let a reader discover. The baseline is trained on 60,000 impressions for 3 epochs and its validation AUC was **still rising at the last epoch**, so these are *undertrained* NRMS numbers — the ablation is valid because every arm has the same budget, but the absolute levels understate a fully-trained NRMS. And + freshness beats the pre-registered + both on both datasets; we report that as an ablation observation rather than promoting it to the headline after the fact, because choosing the winner after seeing the results is precisely the error A1's Finding 5 records.

5. Q4 — serving: memory, p99, and cost per 1,000 queries

Measured per **single user request** ($K = 100$), single process, **with nothing else running** — NRMS training was suspended for the measurement, because A1's error log already contains a conclusion that had to be withdrawn after being taken on a contended machine.

p50 / p99 per stage	MIND	EB-NeRD
retrieve top-100	9.0 / 23.7 ms	9.6 / 22.2 ms
build features	223.0 / 348.0 ms	439.0 / 2,352 ms
score 100 candidates	3.2 / 9.0 ms	4.3 / 6.9 ms
total	237 / 360 ms	452 / 2,369 ms
footprint · throughput · cost/1,000 q · cores for 1,000 QPS	283 MB · 4.2 q/s · \$0.0027 · 239	475 MB · 1.5 q/s · \$0.0077 · 689
p99 < 100 ms SLA	not met	not met

Footprint detail: embeddings 100/193 MB, exposure index 71/123 MB, user vectors 77/61 MB, BM25 19/22 MB, EB-NeRD's session table 44 MB. Start-up (context build) 5.2 s, or 137 s at leaderboard scale.

Little's law check. With one request in flight $X \leq 1/R$: MIND's $R = 0.237$ s gives $X \leq 4.2$ q/s, exactly the measured per-core throughput — so the only way past it is to cut R or add concurrency.

We fail the SLA for an architectural reason, not an algorithmic one.

Stage 1 (9 ms) and stage 3 (3–4 ms) are inside budget; **feature assembly is 93–97% of the request**, because we call a *batch* builder once per request:

- Only **13.5% (MIND) / 16.8% (EB-NeRD)** of that stage is per-user profile work — three decayed profiles, three category-share tables, the BM25 query — measured separately. So "precompute user profiles nightly" is necessary but **not** sufficient; it buys ~30 ms of 223 ms.
- The remainder is fixed per-call cost: DataFrame joins against whole-dataset tables (a 1.2M-row session table, 125k articles, a 65k-entry category map) to answer a 100-row question. The same code processes **~11,500 rows/s in bulk** (2.25M rows in 195 s) and **~450 rows/s one request at a time** — a ~26× penalty that is all overhead.

What would meet p99 < 100 ms (a projection, not a measurement): precompute user profiles into a key-value store (−30 ms), replace per-request joins with array lookups keyed by article row (the targets are static within an hour), and cache article features by (article, hour) — the structure §8 also wants at 10×. Stages 1 and 3 sum to ~13 ms, so a ~20 ms lookup path puts p99 near 40 ms and cost near \$0.0003 per 1,000. **We did not build it**, and saying so beats quoting a number we did not measure. In batch the same system is cheap and fast: 206M rows scored in 155 min, ~22,000 rows/s. It is a strong batch scorer and a weak online one.

6. Q5 — extended evaluation, sliced

All seven metrics, over the supplied-inview pipeline on the local test split, with 1,000-resample percentile CIs. Slices are D26's: cold = history ≤ 5 articles; head/tail by exposure, where an impression is head only if *every* clicked article is in the head set (2,922 MIND / 243 EB-NeRD impressions are mixed and counted in neither, rather than silently assigned).

	MIND: semantic (A1) → LambdaRank	EB-NeRD: semantic (A1) → LambdaRank
all	0.6097 → 0.6144, paired +0.0048 [0.0012, 0.0080]	0.5457 → 0.7428, +0.1971 [0.1953, 0.1989]
cold (n = 3,904 / 680)	0.5642 → 0.5529, −0.0113	0.5538 → 0.7348, +0.1810
warm (n = 18,043 / 186,041)	0.6195 → 0.6277, +0.0083	0.5457 → 0.7428, +0.1971
head-exposure (n = 13,108 / 98,719)	0.5955 → 0.5828, −0.0127 [−0.0176, −0.0079]	0.5565 → 0.7745, +0.2179
tail-exposure (n = 5,917 / 87,759)	0.6319 → 0.6671, +0.0353	0.5334 → 0.7076, +0.1741
diversity (category, K=10)	0.8166 → 0.7352	0.7993 → 0.8029
novelty (bits)	17.34 → 17.69	18.87 → 19.06
coverage	0.0283 → 0.0318	0.0260 → 0.0247

Three readings we would defend:

1. **On EB-NeRD the gain is not an artefact of one population** — it holds in every slice, largest on head-exposure (+0.218) and smallest on tail (+0.174). A model leaning on popularity should be expected to do *better* where articles are widely shown, and it does, which is a consistency check rather than a surprise.
2. **On MIND the aggregate +0.0048 hides two sign flips, which is exactly what slicing is for.** The model is *worse* on cold users (−0.0113) and *worse* on head-exposure impressions (−0.0127 [−0.0176, −0.0079]), and better on warm (+0.0083) and much better on tail-exposure (+0.0353). The mechanism is measurable rather than guessed: on MIND a candidate's own exposure share predicts clicks in the **inverse** direction within its own rack (AUC 0.476 taken positively, 0.524 inverted), so the model learns to down-weight heavily-shown articles. That helps where the clicked article is obscure and hurts where it is the popular one. A cold user additionally has no decayed profile at all, so nothing compensates. Together these argue for a cold-start fallback and for rack-relative rather than absolute popularity features — the improvement we would make next, and the one §4's GBDT property predicts.
3. **The re-ranker is not free on diversity.** On MIND it trades category diversity (0.817 → 0.735) for novelty and coverage; on EB-NeRD it costs coverage (0.0260 → 0.0247) while raising novelty. Coverage carries no CI

(D27): the bootstrap is biased downward for a union statistic, so the raw spread sits in the CSV under its own column names.

7. Q9 — anti-gaming: leakage test, and the price of a serving-time feature

The test (Q9.2). Two properties, not a checklist: future deletion and label blindness (§2), mutation-verified against four planted leaks. A quarantined future feature *fails* the same check, which is how we know the check can see a leak at all.

The ablation (Q9.1). The shipped model is trained twice: on the allowlist, and on the allowlist plus everything in `features/unavailable.py` (`read_time`, `scroll_percentage`, `total_pageviews`, `future_exposure_share_24h`).

AUC	MIND (21,947)	EB-NeRD (186,721)
honest — what ships, and what both leaderboard files came from	0.6144 [0.6104, 0.6185]	0.7428 [0.7415, 0.7441]
+ serving-unavailable	0.5994 [0.5952, 0.6033]	0.7590 [0.7578, 0.7602]
paired difference	−0.0150 [−0.0181, −0.0118]	+0.0162 [0.0153, 0.0172]

The price of cheating on EB-NeRD is +0.016 AUC: real, but under a tenth of what the honest behavioural features bought (+0.197). The two columns that carry it are `total_pageviews` (17% of the leaky model's gain) and `future_exposure_share_24h` (16%).

`read_time` and `scroll_percentage` — the fields the schema most invites you to misuse — are inert here, for a structural reason. EB-NeRD records them per impression, so they are constant across a rack, and a ranking objective only compares candidates *within* a rack; neither reaches the leaky model's top five. **The dangerous leak is the one that varies per candidate**, which is why `total_pageviews` — a per-article whole-dataset aggregate — is the biggest offender here.

Cheating made MIND worse, and the reason is mechanical, not moral. MIND ships no dwell or pageview fields, so the only quarantined column with data is the forward exposure window — and MIND's log ends 2019-11-15 23:58, so that window is 37–138 h wide in training but only 5.7–11.4 h at test time, inflating the feature's mean $3.2\times$ ($0.075 \rightarrow 0.241$). The leaky model leaned on it (118 trees against 18; 15.7% of its gain) and was then handed a differently-distributed input. **So MIND cannot price the leak**; EB-NeRD, where all four columns exist and three are not window-dependent, is where that price is measurable.

8. Where it breaks at 10×

The honest basis for this section is that we **already ran the largest thing we have**: 205,925,868 candidate rows (EB-NeRD) and 93,115,001 (MIND) were featurised and scored locally, in **155 min** and **26 min**, at a 10.6 GB peak on an 11 GB machine. So what follows extrapolates from a measured run rather than from an estimate.

Three things that made the current scale run. (1) *Never materialise the exploded candidate list as strings*: 206M (impression, article) pairs cost ~10 GB as strings and 1.4 GB as one sorted `int64` key per pair. (2) *A deep slice of a lazy frame is not free once a row index is attached*: one 5,000-row chunk at offset 5M cost **297 s**, fixed by writing the prepared frame to Parquet first. (3) *Partition*

by user, not by file order: per-user work (~3.4 ms) was being repeated in every chunk a user appeared in; hashing users into 64 groups pays it once, 2.7 min per 215k-impression group.

At 10× (135M impressions, ~2 G candidate rows), in the order things break:

- **The exposure index, first.** 2 G `int64` keys is **16 GB** — it stops fitting. It must become either time-bucketed shards or a count keyed by (article, hour) instead of by impression, which is a bounded-size structure: 125k articles × 168 hours is 21M cells, not 2 G.
- **Per-user feature building, second.** It is linear and precomputed nowhere: ~8 M users × 3.4 ms ≈ **7.5 h** of profile building alone. A production system would move it to a nightly batch, because a user profile is a function of history, not of the request.
- **Wall clock, third.** ~26 h on this machine, which is past any deadline — but the work is embarrassingly parallel by user group, so it is a horizontal-scaling problem rather than an algorithmic one. Nothing in the pipeline requires a global shuffle.
- **What does *not* break:** the model (322 trees, 1.6 MB), the embeddings (190 MB), and BM25 (a fixed sparse matrix). Stage 1 and stage 2 both stay within a single machine's memory at 10×; it is the *log-derived* features that grow with traffic.

One caveat about numbers taken under pressure. Profiling one leaderboard chunk showed `searchsorted` at 0.7 s per call; the same call on a quiet machine takes **10 ms**. The difference was memory pressure, not the array. A1's error log already contains a conclusion that had to be withdrawn for exactly this reason, which is why the latency figures in §5 were taken with nothing else running.

9. The external check: both leaderboards

	A1's submitted system	A2 (this system)	our offline test AUC
MIND	0.6191	0.6181 (rank 69)	0.6144
EB-NeRD	0.5396	0.7397 (rank 77)	0.7428

EB-NeRD gains +0.2001 AUC over A1, and our offline harness predicted that level to within 0.0031 — on 13.5M unseen impressions, scored by the organisers' own program. That is the strongest evidence we have that the offline numbers in this note mean something.

On MIND the two systems are indistinguishable (−0.0010), although offline we measured +0.0048. We take that at face value rather than explaining it away: the MIND gain was marginal to begin with, its confidence interval [0.0012, 0.0080] was narrow but small, and §6 already showed the aggregate concealing a **−0.0127** loss on head-exposure impressions — which is most of the leaderboard. The offline *level* transferred on both datasets (≤ 0.004); the offline *delta* transferred only where the signal was large.

10. Reproducing, and what is ours

`README.md` carries the one-command rebuild. `AI_USAGE.md` marks every file as AI-generated, AI-generated-then-edited or human-written, and `reports/ai_transcripts/` holds the prompt log (Q7.4). The Codabench worker tooling under `codabench/` is infrastructure, not pipeline: the organisers' scoring program, hidden reference data and metrics are used unmodified.